Purpose

This project aims to digitize and streamline faculty examination and evaluation processes traditionally handled manually. The system focuses on improving task tracking, communication, and scheduling within educational institutions.

By integrating modules like practical code evaluation, task management, meeting scheduling, leave handling, and reporting, the system enhances operational efficiency, transparency, and accountability. It reduces paperwork, minimizes human errors, and provides real-time insights for better decision-making in university examination management.

SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

ACUMEN

AI-Driven Code Understanding & Monitored Examination Network

Project Code - SOCCA-18

Sr. No.	Name	Affiliation	Project Mentor
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Results & Discussion

The system demonstrated significant improvement in task management efficiency and communication between admin and faculty. Task tracking became more transparent, and delays were reduced due to real-time updates. Leave management and meeting scheduling were streamlined, minimizing conflicts.

The system proved reliable, user-friendly, and effective in handling multiple administrative operations simultaneously. Compared to traditional methods, it showed better accuracy, reduced workload, and enhanced productivity in faculty supervision and practical examination evaluation.

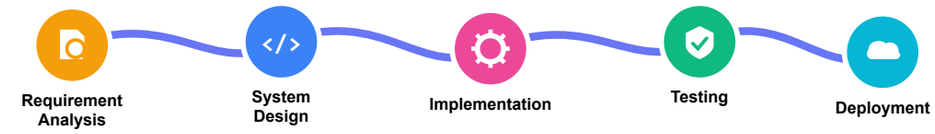
Abstract

The **"ACUMEN"** (AI-Driven Code Understanding & Monitored Examination Network) system is a web-based solution designed to replace traditional manual record-keeping and subjective grading methods. It enables administrators to manage faculty, assign examination tasks, schedule practical lab sessions, and monitor student performance through a centralized platform.

Faculty members can update task status, verify multi-tier code submissions, apply for leave, and access real-time notifications efficiently. Powered by AST construct decomposition and AI analysis, student code is evaluated objectively on logic rather than code length. The system ensures better coordination, reduces delays, and improves productivity. It is scalable, user-friendly, and suitable for modern educational environments aiming for digital transformation.

Methodology

- **Requirement Analysis:** Identified problems in manual faculty management and examination evaluation.
- **System Design:** Designed Admin, Faculty, and Student examination modules.
- **Development:** Built using modern web technologies (React, Node.js, Express, MongoDB Atlas).
- **Implementation:** Integrated features like task lifecycle, AST logic scoring, leave approval, and notifications.
- **Testing:** Conducted functional, anti-cheating tab detection, and algorithmic usability testing.
- **Deployment:** Cloudflare Pages and cloud API cluster for secure, real-time access.



Conclusion & Future Scope

The "ACUMEN" system successfully provides a digital solution for faculty and examination management, improving efficiency and reducing manual workload. It ensures better coordination and transparency in institutional operations.

Future enhancements may include:

- AI-based task recommendations
- Mobile application integration
- Biometric attendance system
- Advanced analytics and reporting
- Cloud-based scalability

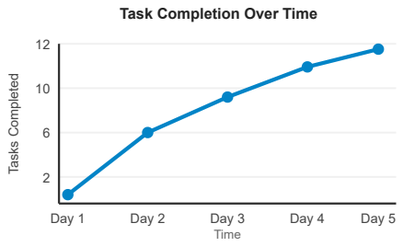
Literature Survey

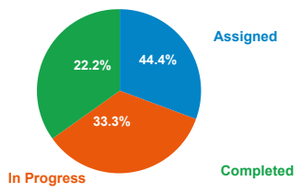
Existing systems for academic evaluation and faculty management often rely on manual processes or partially digital solutions, leading to inefficiencies and lack of real-time updates. Research highlights the importance of automation in improving administrative workflows and reducing redundancy in higher education institutions.

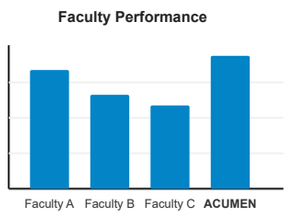
Various studies suggest that centralized management systems enhance communication and decision-making. However, many systems lack integration of multiple modules such as automated practical code tracking, anti-cheating proctoring, leave management, and granular analytics. This project bridges that gap by providing a unified, efficient, and scalable solution.

Data Analysis (Data / Tables / Graphs / Images)

Status	Count
Assigned	**20**
In Progress	**15**
Completed	**10**
Verified	**5**







References

[1] A. Kumar and R. Singh, "Smart Faculty Management System," IEEE Xplore, 2022.

[2] S. Patel et al., "Automation Techniques in Educational Administration," International Journal of Engineering Research & Technology (IJERT), 2021.

SDGs Covered

